Graphs 5 — Inequalities on graphs · Perpendicular lines · Mid-points · Distances

Squared paper needed. Shade the regions *not* wanted, as in the exam. Marks in brackets. ★ = harder question.

1 Illustrate the inequality $-2 < x \leq 4$ on a graph, shading the unwanted regions. [2]

2 [8]

a Draw axes from 0 to 10 and on them draw the lines $x = 5$, $y = 7$ and $2x + 3y = 24$.

b Shade the regions not satisfied by $x \leq 5$, $y \leq 7$, $2x + 3y \leq 24$, $x \geq 0$, $y \geq 0$.

c Write down the point with integer coordinates in the unshaded region that is furthest from the origin.

3 A is $(2,\ -3)$ and B is $(6,\ 5)$. [7]

a Find the length of AB.

b Find the mid-point of AB.

c Find the equation of the perpendicular bisector of AB.

4 A is $(1,\ -4)$ and B is $(7,\ 4)$. AB is a diameter of a circle with centre C. [8]

a Find the coordinates of C.

b Find the radius of the circle.

c Show that the point $D(8,\ 3)$ lies on the circle.

d A circle theorem says that AD is perpendicular to BD. Show that this is true.

5 Find the equation of the line perpendicular to $y = 3x - 2$ that passes through $(6,\ 1)$. [3]

Total: 28 marks

Check: solid line for $\leq$ / $\geq$, dotted for $<$ / $>$ · test a point to decide which side · perpendicular gradients multiply to -1 · mid-point = mean of the coordinates · distance = Pythagoras on the coordinate differences.